

Thesis Inventory System

Software Requirements Specification (SRS)

Version 1.0

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Thesis Inventory System. The system is a web based catalog that stores and presents key information about completed thesis papers so that students, faculty, and researchers can browse and search past work for reference. This document describes the system scope, user roles, functional requirements, data requirements, interface requirements, and nonfunctional requirements that will guide design, development, and evaluation.

1.2 Document Conventions

Functional requirements are written as shall statements and are assigned identifiers in the form FR-X.Y for traceability. The word system refers to the Thesis Inventory System described in this document. The terms Administrator, Department, and Viewer refer to the three user classes defined in Section 2.3.

1.3 Intended Audience

This document is intended for the project proponents, the thesis adviser and panel, and any developers or evaluators involved in building or reviewing the system. Readers are assumed to have a basic understanding of web applications and database driven systems.

1.4 Project Scope

The system provides a centralized, searchable inventory of thesis records. For each thesis it stores descriptive information only: the title, the authors, the year, the program or department, a short description (abstract), the recommendations of the study, the adviser, the panelists, and keywords. The system does not store or distribute the full thesis document. The intent is to let users read a thesis summary and its recommendations to gain ideas and direction for their own research, without granting access to the complete paper.

The system supports three user classes. Administrators manage department accounts and oversee the system. Department users create and maintain the thesis records that belong to their own department. Viewers are members of the public who can browse and search all records without logging in. The system also keeps an activity log of important actions for accountability.

1.5 Definitions, Acronyms, and Abbreviations

SRS - Software Requirements Specification.

CRUD - Create, Read, Update, and Delete operations on records.

RBAC - Role Based Access Control, the model used to grant permissions by user role.

OCR - Optical Character Recognition, the process of extracting text from an image of a printed page.

Thesis Record - A single entry in the inventory containing the descriptive fields of one thesis.

Department - An academic unit and also the user account type responsible for that unit's thesis records.

2. Overall Description

2.1 Product Perspective

The Thesis Inventory System is a new, self contained web application. It is not a replacement for an existing system and does not depend on integration with external institutional systems. It is built as a server rendered web application with role based access, a relational database for storage, and an optional camera based text capture feature to reduce manual encoding.

2.2 Product Functions

At a high level the system allows the following:

- Administrators to create and manage department accounts, configure permissions, and review the activity log.
- Department users to add, edit, and remove thesis records that belong only to their own department.
- Department users to capture the abstract and recommendations from a printed copy using a device camera, then review and correct the text before saving.
- Viewers to browse and search all thesis records across all departments without an account.
- The system to record an activity log of account creation and of thesis create, edit, and delete actions.

2.3 User Classes and Characteristics

The system defines three user classes with distinct privileges:

Administrator A trusted system manager. Creates department accounts, assigns and manages permissions, and views the activity log. When deleting a department account, the administrator decides whether the department's thesis records are kept or removed. The administrator does not normally encode thesis content.

Department A faculty or staff account that represents one academic department. Creates, edits, and removes thesis records, but only for records that belong to its own department. May use the camera capture feature when encoding records. Cannot manage other departments or view administrative functions.

Viewer Any member of the public. Requires no account and does not log in. Can browse and search across all thesis records and read the descriptive fields, but cannot create, edit, or delete anything.

2.4 Operating Environment

The system is a web application accessed through a standard modern web browser on desktop or mobile devices. The camera capture feature requires a device with a camera and a browser that permits camera or file access. The application is built on the Laravel framework (PHP), uses

Blade templates for the server rendered frontend, and uses Supabase, a hosted PostgreSQL database, for storage. It is served from a standard web server.

2.5 Design and Implementation Constraints

- The system shall be developed using the Laravel framework (PHP) and shall use Supabase (hosted PostgreSQL) as its relational database. Laravel shall connect to the Supabase database directly and manage data access through Eloquent and migrations.
- Authentication and role based access control shall be handled within Laravel using an established permission package (for example, spatie/laravel-permission), rather than through Supabase Auth.
- The activity log shall be implemented using an established Laravel activity logging package (for example, spatie/laravel-activitylog).
- Text capture shall use an OCR engine. The client side option is Tesseract.js running in the browser. A server side option is a Tesseract wrapper for PHP, or a cloud vision service. OCR output is always reviewable and editable before saving.
- The frontend shall be implemented using Laravel Blade templates (server rendered). Client side interactions, such as adding or removing multi value input boxes and the camera based text capture, shall be handled with lightweight JavaScript (for example, Alpine.js together with Tesseract.js).
- The system shall store descriptive metadata only. It shall not store, host, or distribute the full thesis document or any uploaded paper file.

2.6 Assumptions and Dependencies

- It is assumed that departments have permission to publish the descriptive information and recommendations of the thesis works they encode.
- It is assumed that printed copies used for camera capture are reasonably clean and legible, since OCR accuracy depends on image quality.
- The system depends on the availability of the chosen OCR engine and the hosting environment for normal operation.

2.7 Technology Stack

The system uses the following technologies:

Layer	Technology
Framework	Laravel (PHP)
Frontend	Laravel Blade templates (server rendered), with Alpine.js for lightweight client side interactions
Database	Supabase (hosted PostgreSQL), accessed by Laravel through Eloquent and migrations

Layer	Technology
Authentication and Roles	Laravel authentication with a permission package (for example, spatie/laravel-permission)
Activity Logging	A Laravel activity logging package (for example, spatie/laravel-activitylog)
Text Capture (OCR)	Tesseract.js in the browser, or a server side Tesseract wrapper or cloud vision service

3. Functional Requirements

3.1 Account and Access Management

FR-1.1 The system shall allow only the Administrator to create department accounts.

FR-1.2 The system shall require department users to log in with valid credentials before accessing department functions.

FR-1.3 The system shall not require Viewers to register or log in; browsing and searching shall be publicly available.

FR-1.4 The system shall enforce role based access so that each user class can access only the functions permitted to it.

FR-1.5 The system shall allow the Administrator to manage permissions and to enable or disable department accounts.

3.2 Administrator Functions

FR-2.1 The system shall allow the Administrator to create, edit, and deactivate department accounts.

FR-2.2 The system shall allow the Administrator to view the activity log of recorded actions.

FR-2.3 When the Administrator deletes a department account, the system shall prompt the Administrator to choose whether to keep or delete that department's thesis records.

FR-2.4 If the Administrator chooses to keep the records, the system shall retain them in the inventory and they shall remain searchable by Viewers.

FR-2.5 If the Administrator chooses to delete the records, the system shall remove that department's thesis records together with the account.

3.3 Department Functions

FR-3.1 The system shall allow a Department user to create a new thesis record.

FR-3.2 The system shall allow a Department user to edit an existing thesis record.

FR-3.3 The system shall allow a Department user to delete a thesis record.

FR-3.4 The system shall restrict each Department user to creating, editing, and deleting only the thesis records that belong to its own department.

FR-3.5 The system shall associate every thesis record with the department account that created it.

FR-3.6 The system shall prevent a Department user from viewing or modifying records owned by other departments through management functions.

3.4 Thesis Record Management

FR-4.1 The system shall store the following fields for each thesis record: title, authors, year, program or department, short description (abstract), recommendations, adviser, panelists, and keywords.

FR-4.2 The system shall allow authors, advisers, panelists, and keywords to hold multiple values, each entered in its own input box, with the ability to add or remove boxes as needed.

FR-4.3 The system shall require at least the title and a short description before a record can be saved.

FR-4.4 The system shall not provide any field for uploading or storing the full thesis document.

FR-4.5 The system shall display each thesis record's descriptive fields to Viewers in a readable layout.

3.5 Camera Text Capture (OCR)

FR-5.1 The system shall allow a Department user to capture text from a printed copy using the device camera or an uploaded image.

FR-5.2 The system shall apply OCR to the captured image and place the extracted text into the relevant field, such as the short description or the recommendations.

FR-5.3 The system shall allow the Department user to review and edit the extracted text before saving the record.

FR-5.4 The system shall allow capture of the abstract and the recommendations separately, each into its own field.

FR-5.5 The camera capture feature shall be optional; the Department user shall always be able to type the information manually instead.

3.6 Browsing and Search (Viewer)

FR-6.1 The system shall allow Viewers to browse thesis records across all departments without logging in.

FR-6.2 The system shall allow Viewers to search thesis records by title, author, adviser, keywords, year, and program or department.

FR-6.3 The system shall return matching results and allow the Viewer to open a record to read its descriptive fields.

FR-6.4 The system shall present search results across all departments in a single combined result set.

3.7 Activity Logging

FR-7.1 The system shall record an activity log entry when a department account is created.

FR-7.2 The system shall record an activity log entry when a thesis record is created, edited, or deleted.

FR-7.3 Each log entry shall include the actor (the account responsible), the action performed, the affected record, and a timestamp.

FR-7.4 The system shall make the activity log viewable by the Administrator only.

4. Data Requirements

4.1 Data Entities

The system manages the following primary data entities:

Entity	Description
User Account	Represents an Administrator or a Department account, including credentials and role.
Department	An academic unit that owns thesis records. Linked to its department account.
Thesis Record	A single thesis entry containing the descriptive fields listed in Section 4.2.
Activity Log	A record of account creation and thesis create, edit, and delete actions.

4.2 Thesis Record Attributes

Each thesis record contains the following attributes:

Attribute	Description
Title	The title of the thesis. Required.
Authors	One or more names of the student researchers. Each name in its own box.
Year	The year the thesis was completed.
Program / Department	The academic program or department the thesis belongs to.
Short Description	A brief abstract summarizing the study. Required.
Recommendations	The recommendations for future research drawn from the study.
Adviser	One or more thesis advisers. Each name in its own box.
Panelists	One or more panel members. Each name in its own box.
Keywords	One or more keywords or tags used to improve search. Each in its own box.
Owning Department	The department account that created and owns the record.

5. External Interface Requirements

5.1 User Interfaces

- A public landing and search interface for Viewers, with a search bar and filterable, browsable results.
- A thesis detail view that presents all descriptive fields of a record in a clear, readable layout.
- A department dashboard with forms to create, edit, and delete thesis records, including add and remove controls for multi value fields and a camera capture control.
- An administrator dashboard to manage department accounts, set permissions, and view the activity log.

5.2 Hardware Interfaces

The system uses the device camera on the Department user's device for the optional text capture feature. No other special hardware is required. Standard desktop or mobile devices with a web browser are sufficient for all roles.

5.3 Software Interfaces

- Supabase (hosted PostgreSQL) as the relational database for storing user accounts, departments, thesis records, and the activity log.
- An OCR engine for text capture, such as Tesseract.js in the browser, a Tesseract wrapper for PHP on the server, or a cloud vision service.
- The Laravel framework and supporting packages for authentication, role based access control, and activity logging.

6. Nonfunctional Requirements

6.1 Security

NFR-1.1 The system shall protect department and administrator functions behind authentication and role based access control.

NFR-1.2 The system shall store account passwords using secure hashing and shall not store passwords in plain text.

NFR-1.3 The system shall ensure that a Department user cannot modify records belonging to another department.

6.2 Performance

NFR-2.1 The system shall return search results for typical queries within a few seconds under normal load.

NFR-2.2 The system shall remain responsive while OCR processing occurs, and shall indicate progress to the user.

6.3 Usability

NFR-3.1 The system shall allow Viewers to search and read records without any training or account.

NFR-3.2 The system shall present clear add and remove controls for multi value fields such as authors and panelists.

NFR-3.3 The system shall let Department users review and correct OCR output before saving.

6.4 Reliability

NFR-4.1 The system shall preserve existing thesis records during normal operation and shall only remove records through explicit, confirmed actions.

NFR-4.2 The system shall confirm destructive actions, such as deleting a record or a department account, before carrying them out.

6.5 Maintainability

NFR-5.1 The system shall be organized using the Laravel framework's standard structure to support maintenance and future enhancements.

NFR-5.2 The system shall use established packages for permissions and activity logging rather than custom implementations where practical.